STACKS, COHOMOLOGY OF SCHEMES (COHERENT.TEX): LEMMA-CLOSED-PROPER-OVER-BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7429-7442 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 7444-7475, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 49 of queue_coherent.json (pb=1672, stmt 7429-7442, proof 7444-7475).

Editorial note. co02 wave: conflict-free remaining accept (old-session co01-co08 build did not cover this label; fresh adjudication 2026-10-14 accepted).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-closed-proper-over-base' also exists in stacks-spaces-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0C4H: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W5: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T3: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W3: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03GN: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09MQ: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T5: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C4J: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01J4: sources/stacks-coherent/pinned-original-source.tex, source lines 7475–7475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $Z \subset X$ be a closed subset. The following are equivalent*

1. *the morphism $Z \to S$ is proper if Z is endowed with the reduced induced closed subscheme structure (Schemes, Definition 01J4),*
2. *for some closed subscheme structure on Z the morphism $Z \to S$ is proper,*
3. *for any closed subscheme structure on Z the morphism $Z \to S$ is proper.*

Proof. The implications (3) $\Rightarrow$ (1) and (1) $\Rightarrow$ (2) are immediate. Thus it suffices to prove that (2) implies (3). We urge the reader to find their own proof of this fact. Let Z' and Z'' be closed subscheme structures on Z such that $Z' \to S$ is proper. We have to show that $Z'' \to S$ is proper. Let $Z''' = Z' \cup Z''$ be the scheme theoretic union, see Morphisms, Definition 0C4H. Then Z''' is another closed subscheme structure on Z. This follows for example from the description of scheme theoretic unions in Morphisms, Lemma 0C4J. Since $Z'' \to Z'''$ is a closed immersion it suffices to prove that $Z''' \to S$ is proper (see Morphisms, Lemmas 01W5 and 01W3). The morphism $Z' \to Z'''$ is a bijective closed immersion and in particular surjective and universally closed. Then the fact that $Z' \to S$ is separated implies that $Z''' \to S$ is separated, see Morphisms, Lemma 09MQ. Moreover $Z''' \to S$ is locally of finite type as $X \to S$ is locally of finite type (Morphisms, Lemmas 01T5 and 01T3). Since $Z' \to S$ is quasi-compact and $Z' \to Z'''$ is a homeomorphism we see that $Z''' \to S$ is quasi-compact. Finally, since $Z' \to S$ is universally closed, we see that the same thing is true for $Z''' \to S$ by Morphisms, Lemma 03GN. This finishes the proof. $\square$